

Inequalities 1

Date 9/9/2026

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→ Meaning: Comparison

Unit be must same for comparison

eg Weight (1) - Weight (2).

Height (1) - Height (2).

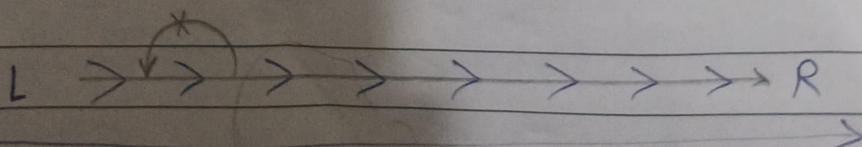
Eg-1) $A > B > C = D$ ① $A > C$ (T)④ $C > D$ (F)② $C > A$ (F)⑤ $C < D$ (F)③ $B = C$ (F)⑥ $A > D$ (T)

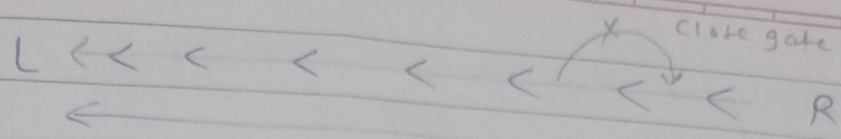
- Rule - There should be relation betⁿ the variables. & If there is no relation it is false

Eg-1) $P > Q > R < S < T$ ① $P > R$ (T)④ $R > T$ (F)② $R > P$ (F)⑤ $S > T$ (F)③ $Q > S$ (F)⑥ $T > S$ (T)2) ~~A~~ $B > C < D = E$ ① $B > D$ (F)④ $D > E$ (F)② $D > B$ (F)⑤ $D < E$ (F)③ $D > C$ (T)

* Gate Method:

(close gate)





Eg-1) $P > Q > R > S > T$

- | | |
|----------------|----------------|
| 1) $P > R$ (T) | 3) $T < R$ (T) |
| 2) $P > T$ (T) | 4) $S < Q$ (T) |

2) $A > B > C < D < E < F$

- | | |
|----------------|----------------|
| 1) $A > C$ (T) | 5) $F > D$ (F) |
| 2) $C < A$ (T) | 6) $D < F$ (T) |
| 3) $C > E$ (F) | 7) $B > E$ (F) |
| 4) $E > C$ (T) | 8) $A > F$ (F) |

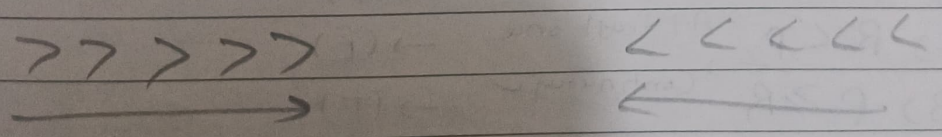
Session 3

* Inequality - 2

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Eg- $A > B > C < D < E$

- | | |
|----------------|----------------|
| 1) $A > C$ (T) | 4) $E > C$ (T) |
| 2) $B > D$ (F) | 5) $C < E$ (T) |
| 3) $D > B$ (F) | |



Eg- $P > Q ; Q > R ; R > S$

- 1) $P > R \rightarrow$ (T)
- 2) $R < P \leftarrow$ (T)
- 3) $P > S \rightarrow$ (T)

$$3) A > B ; C < B > D , C > E$$

$$1) A > C \text{ (T)} \quad 5) E > A \text{ (F)}$$

$$2) C > D \text{ (F)}$$

$$3) A > E \text{ (T)} \quad A \rightarrow B \rightarrow B \rightarrow C \rightarrow E$$

$$4) B > E \text{ (T)} \quad B \rightarrow C \rightarrow E$$

* $> < =$ } Basic symbol

$> + = \rightarrow \geq$
 $< + = \rightarrow \leq$ } Composite symbol

• Note:

$$\begin{array}{c} A \rightarrow B \\ A > B > C \end{array}$$

At least once $\textcircled{1} A > C \text{ (T)}$

$\textcircled{2} A \geq C \text{ (F)}$ continuous

Eg-1) $P \geq Q \geq R$

Imp

$$1) P > R \xrightarrow{\text{At least once}} \rightarrow (F)$$

$$2) R < P \xrightarrow{\text{At least once}} \rightarrow (F)$$

$$3) P \geq R \xrightarrow{\text{Continuous}} \rightarrow (T)$$

$$2) P \geq Q \geq R > S$$

$$1) P > R \text{ F}$$

$$4) P \geq S \text{ F}$$

$$2) R < P \text{ F}$$

$$5) P > S \text{ T}$$

$$3) P \geq R \text{ T}$$

3) $P \geq Q \leq R \geq S > T > U \geq W$

- | | | | | | |
|---------------|---|---------------|---|------------|---|
| 1) $P > Q$ | F | 5) $R > U$ | T | 9) $T > W$ | T |
| 2) $P > R$ | F | 6) $R > W$ | F | | |
| 3) $R > T$ | F | 7) $R > S$ | T | | |
| 4) $R \geq T$ | T | 8) $T \geq W$ | F | | |

Session-4

10/9/24

* Inequality-3

Eg-1) $P \geq Q \geq R < S \geq T > U$

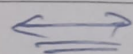
- | | | | |
|---------------|---|---------------|--------------|
| 1) $P \geq R$ | T | 5) $S > T$ | F |
| 2) $P > R$ | F | 6) $S > U$ | T |
| 3) $Q > S$ | F | 7) $U > S$ | F |
| 4) $S \geq U$ | F | 8) $T \leq S$ | T |

• Note:-

$(> , <) \rightarrow$ At least once

$(\geq , \leq) \rightarrow$ continuous

$(=) \rightarrow$ Left to right and right to left



Eg-1) $A \geq B = C > D \leq E = F > G$

- | | | | |
|---------------|---|---------------|---|
| 1) $A > C$ | F | 5) $E \geq G$ | F |
| 2) $A \geq C$ | T | 6) $E > F$ | F |
| 3) $B > D$ | T | 7) $D = E$ | F |
| 4) $B > E$ | F | 8) $C \geq D$ | F |

* Exam - level - Questions

Q-1)

1) Statement: $x > y \geq z$

a) $z > y$ (F) b) $y = z$ (F)

2) Statement: $D \geq E > F = G$

a) $D > G$ (T) b) $F > G$ (F)

3) Statement: $B > J \geq R > Z$

a) $J > Z$ (T) b) $B > Z$ (T)

4) $A > B = C < D$

a) $A < D$ (F) b) $B > D$ (F)

5) Statement - $A > B < C \leq D$

a) $D > A$ (F) b) $B < D$ (T)

Q-2)

1) Statement: $M \leq K > L = Y, P \leq T > M$

i) $P > Y$ (F) ii) $T < L$ (F)

2) Statement: $Z > T, T < M, M < J$

i) $T < J$ (T) ii) $J < Z$ (F)

3) Statement :

$$P = Z, C \geq G, G \geq Q, Q \geq R, T \geq C$$

i) $G \geq Z$ (T) ii) $C \geq R$ (T)

4) Statement : $A > B > C, D > E > F, D > C$

i) $E > C$ (F) ii) $F > B$ (F)

5) Statement : $K < M, M \geq N, N > O$

i) $O < M$ (T) ii) $O < K$ (F)

Session-5

* Hr-W - Prelims Q. 1-25. Solve 11/9/26

9. Directions (Q. 1-25) : In these questions relations between different elements is shown in the statements. These statements are followed by two conclusions. Give Answer

- A. Only ~~Conclusion I~~ Conclusion I follows
- B. only Conclusion II follows
- C. Either Conclusion I or II follows
- D. Neither Conclusion I or II follows
- E. Both Conclusion I and II follows.

1) Statement : $X > Y \geq Z$

I) $Z > Y$ (F) II) $Y = Z$ (F) $\Rightarrow D$

2) Statement : $D \geq E > F = G$

I) $D > G$ (T) II) $F > G$ (F) $\Rightarrow A$

$$p > c > q$$

$$c < d$$

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3) Statement: $B > J \geq R > Z$
I) $J > Z$ (T) II) $B > Z$ (T) $\Rightarrow E$

4) Statement: $A > B = C < D$
I) $A < D$ (F) II) $B > D$ (F) $\Rightarrow D$

5) Statement: $A > B < C \leq D$
I) $D > A$ (F) II) $B < D$ (T) $\Rightarrow B$

6) Statement: $M \leq K > L = Y, P \leq T > M$
I) $P > Y$ (F) II) $T < L$ (F) $\Rightarrow D$

7) Statement: $Z > T, T < M, M \leq J$
I) $T < J$ (T) II) $J < Z$ (F) $\Rightarrow A$

8) Statement: $Q = Z, C \geq G, G \geq Q, Q \geq R, J \geq C$
I) $G \geq Z$ (T) II) $C \geq R$ (T) $\Rightarrow E$

9) Statement: $A > B > C, D > E > F, D > C$
I) $E > C$ (F) II) $F > B$ (F) $\Rightarrow D$

10) Statement: $K < M, M \geq N, N > O$
I) $O < M$ (T) II) $O < K$

11) Statement: $A > B \geq C < D, C = E > G$
I) $D > E$ (T) II) $B > E$ (F) $\Rightarrow A$

12) Statement: $P \leq Q > M \geq N, Q = S$
I) $S > P$ (F) II) $N < S$ (T) $\Rightarrow B$

$$W \leq L \quad U > T = V \geq S$$

13) Statements: $S > M = Z > T < Q > V$

I) $V = S$ (F) II) $Q > M$ (F) $\Rightarrow D$

14) Statements: $T < U = V \geq S > P \geq Q$

I) $S > T$ (F) II) $V \geq Q$ (F) $\Rightarrow A$

15) Statements: $M \geq N > R > W, E = J < L \geq W$

I) $E > W$ (T) II) $M > L$ (F) $\Rightarrow A$

16) Statements: $S \leq L \leq I = P \geq E > R, L > Q$

I) $P \geq S$ (T) II) $I > R$ (T) $\Rightarrow E$

17) Statement: $G > R \geq E = A \leq T \leq S, D \leq A \leq J$

I) $T \geq D$ (F) II) $R > S$ (F) $\Rightarrow D$

18) Statements: $A \geq B > C \leq D \leq E < F$

I) $A \geq E$ (F) II) $C < F$ (T) $\Rightarrow B$

19) Statements: $G > R \geq E = A \leq T \leq S, D \leq A \leq J$

I) $J > G$ (F) II) $J = E$ (F)

20) Statements: $S \leq L \leq I = P \geq E > R, L > Q$

I) $L < R$ (F) II) $E \geq Q$ (F)

21) Statements: $Q \geq P > K, P > F < C, A > W$

I) $Q > F$ (F) II) $A > C$ (F)

22) Statements: $Q > V < B \leq P, A \leq M \leq U, G < M$

I) $P > U$ (F) II) $A < Q$ (T)

$L > P$ $J \leq A$

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23) $L > P, P < N, N \geq A$

I) $A < P$ (T) II) $L > N$ (F)

24) $K \geq T \geq J, R < L, L > O$

I) $O > T$ (F) II) $K > L$ (F)

25) $J \geq A > D = E, L < A < M$

I) $M < J$ (F) II) $J > L$ (T)

9. 1) $S > M \geq D > H \leq R \leq T < W$

I) $S > H$ (T)

II) $W > H$ (T)

III) $R < W$ (T)

IV) $M > T$ (F)

2) $W \geq A, P = D, F < E, W < P, W < C, D \geq E$

I) $D < E$ (F)

II) $A < P$ (T)

III) $A > P$ (F)

IV) $E < C$ (F)

3) $J \geq A > D = E, L < A < M$

I) $M < J$ (F)

II) $J > L$ (T)

III) $A > L$ (F)

IV) $E < M$ (T)

4) $Y > F \leq O \leq P, F \geq U < T$

I) $Y > P$ (F)

II) $T < F$ (F)

III) $O > T$ (F)

IV) $P < U$ (F)

5) $M > H \leq Y \leq R < U = Z \geq E$

I) $M > R$ (F)

II) $Z \leq R$ (F)

III) $R > E$ (F)

IV) $Z > H$ (T)

6) $P \geq Q, R > M, P \geq O, Q \geq R, N = Q$

I) $R > O$ F

II) $O > Q$ F

III) $Q > M$ T

IV) $N > P$ F

7) $A \geq B, C = B, E > F, A \leq D, B \geq F$

I) $C < E$ F

II) $D \geq B$ T

III) $A \geq F$ T

IV) $E > D$ F

8) $P > Q, S \geq U, Q \leq T, R = S, R < T$

I) $P \geq U$ F

II) $U < T$ T

III) $A \leq P$ F

IV) $P < T$ F

9) $A = B, C \leq D, E > C, B < A$

I) $D < B$ F

II) $D > E$ F

III) $A > E$ F

IV) $A > C$ F

10) $Q \geq P, R = S, U < T, Q < R, Q < V, S \geq T$

I) $S < U$ F

II) $P > R$ F

III) $T < V$ F

IV) $P < R$ T